

6-Week Execution Plan

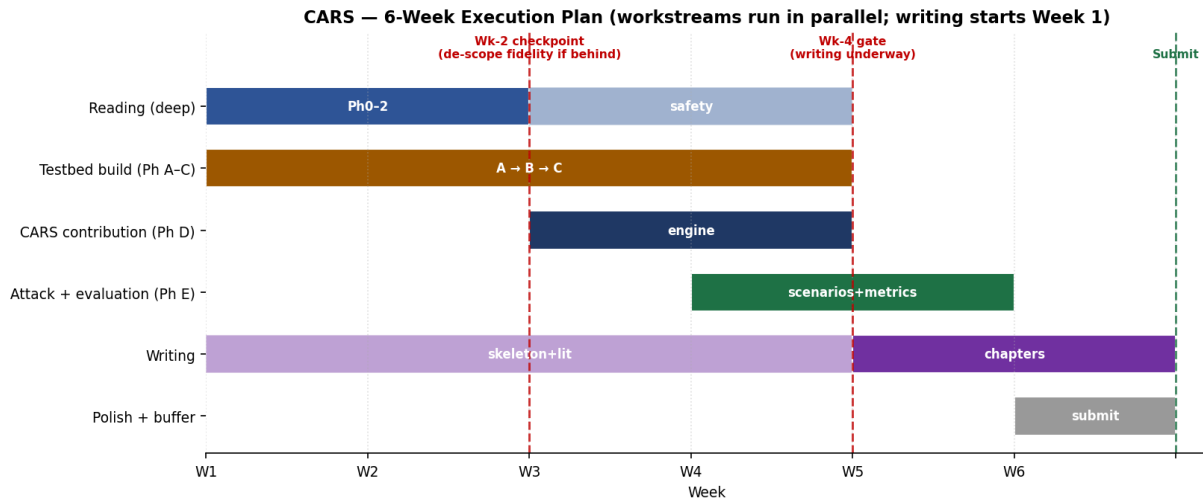
Project CARS — Reactive SDN for Securing ICS (working title)

Indicative dates Wk1 = Mon 6 Jul 2026 → submit ~16 Aug 2026 · anchor to your real deadline

1. Operating principles

- **Parallel streams, not phases-in-series.** Build, read, and write run concurrently every week. Writing starts Week 1 — never left to the end.
- **De-scope fidelity, never the contribution.** If behind, cut testbed richness via the ladder in §6. The CARS engine, the evaluation, and the write-up are protected.
- **Resolve the crux early.** CC-1 (conduit vs intra-zone scope) and CC-3 (how safety is proven) are settled in Week 1, before the engine is coded. CC-4 (does CARS add latency) is measured in Week 2.
- **Build outward from a working core.** Minimal reactive loop first (Week 1–2); fidelity layered on only once it works.

2. Timeline at a glance



Week	Build	Read / Decide	Write
W1	Devices ready; Phase A: Factory I/O + real PLC + Ignition HMI on flat OVS	Ph 0–2 deep; LOCK contribution; settle CC-1 & CC-3	Skeleton + Intro + start Background
W2	Phase B (Ryu ↔ OVS; measure latency=CC-4); Ph C-lite (IDS→Event API; manual trigger→flow); historian+OPC UA	Safety cluster (Ph 8)	Methodology + Testbed chapter
W3	Phase D: CARS engine — criticality model + safety-constrained response + audit log	Taper (policy-checker, twin-IR)	Finish Lit Review; Design chapter
W4	Zones in GNS3 (pfSense×2, DMZ, corp) as time allows; Ph E attack suite; first metrics	—	GATE: Intro+Bg+Method drafted; Implementation ch.
W5	Ph E full run: scenario suite +	—	Results + Analysis;

Week	Build	Read / Decide	Write
	operator metrics; analysis		Discussion
W6	Code freeze (bugfix only)	Cite-time checks (CHAOS id, NIST r3)	Conclusion, Abstract, polish, SUBMIT

3. Week-by-week detail

Week 1 — Foundations & the minimal loop

- Complete the pre-build checklist; Phase A: PLC controls the Factory I/O process (S7 driver), visible on the Ignition HMI, all traffic through Open vSwitch on a single flat network.
- **Decide & record:** lock the single contribution; resolve CC-1 (does CARS act only at conduits or also intra-zone?) and CC-3 (safety proven by envelope-invariant check and/or twin dry-run?). Confirm D4/D5.
- **Exit:** working process-under-PLC-control through OVS; contribution + safety method locked; dissertation skeleton + Intro drafted.

Week 2 — SDN baseline, detection & reactive skeleton

- Phase B: Ryu manages OVS; measure baseline control-loop latency/jitter (CC-4). Phase C-lite: Suricata/Zeek on an OVS mirror emit alerts to the Event API; a manual trigger makes Ryu install one block/redirect rule. Stand up historian + OPC UA.
- **CHECKPOINT (end W2):** is the alert→controller→flow-change loop solid AND is the added latency acceptable? If no → trigger the de-scope ladder now.

Week 3 — CARS engine (the contribution)

- Phase D: criticality model (tag the PLC control loop vs unknown hosts); safety-constrained response (allow / block / mirror / redirect); safety guard (never hard-block the critical loop — mirror/redirect instead; policy/twin check before install); reversibility + machine-readable audit log.
- **Exit:** unseen device that alerts is auto-blocked; alert on the critical loop is redirected; Factory I/O process never interrupted; every action logged.

Week 4 — Zones, attack suite & first evaluation

- Build the virtual zones in GNS3 (two pfSense firewalls, DMZ, corporate/Kali) as far as time allows; implement the attack suite (recon, unauthorized Modbus/S7 write, false-data injection, DoS; full IT→OT kill chain if zones are ready); first metrics run.
- **GATE (end W4):** writing must be underway — Introduction, Background and Methodology drafted. If not, stop building and write.

Week 5 — Full evaluation & results

- Phase E complete: run the full scenario suite; collect operator-relevant metrics — zero defence-induced process trips, false-block rate on critical flows, mean-time-to-mitigate, controller load, audit-trail completeness; analyse; sensitivity checks if time. Add Tier-3 fidelity only if ahead.
- **Exit:** complete, reproducible results table per scenario; analysis written.

Week 6 — Writing, polish, buffer, submit

- Freeze code (bugfixes only). Write Discussion (incl. deployability / product argument + limitations), Conclusion, Abstract. Final citation verification (CHAOS article ID; cite NIST SP 800-82 Rev. 3; confirm co-engineering survey authors). Diagrams, proofread, format; leave buffer for supervisor review. Submit.

4. Checkpoints & gates

When	Gate	If it fails
End W1	Minimal loop (PLC ↔ Factory I/O ↔ OVS) up; contribution locked	Extend into W2; cut nothing yet but flag risk
End W2	Reactive loop solid + latency acceptable (CC-4)	Trigger de-scope ladder (§6) immediately
End W4	Writing underway (Intro/Bg/Method drafted)	Pause building; writing takes priority
End W5	Full results captured	Freeze scope; write up what exists honestly
~W6	Submit	—

5. Definition of a strong submission

- A working CARS engine on the testbed (Tier 3 if achieved; Tier 2 acceptable) with a real S7-1200/1500 and a process that can visibly go unsafe.
- ≥4–5 attack scenarios evaluated with operator-relevant metrics — crucially, zero defence-induced process trips demonstrated.
- Positioned against the key literature including the safety cluster (Piedrahita, process-aware review, TRITON, IEC 62443).
- An explicit deployability / product argument (zone-boundary overlay, brownfield-friendly).
- Verified citations; word count per your programme handbook (confirm the exact limit).

6. De-scope ladder (cut in this order if behind)

Cut fidelity top-down until the schedule is recoverable. Each cut is honestly reported in the write-up as a scope decision + future work.

1. Corporate/IT zone + dual firewalls → single IT/OT boundary.
2. Industrial DMZ (historian replica + jump host).
3. Second soft-PLC (OpenPLC).
4. Extra protocols (OPC UA / MQTT) → keep Modbus + S7comm.
5. Full IT→OT kill chain → intra-OT attacks only.

Core that must survive every cut: real PLC + Factory I/O + OVS + Ryu + one IDS + the CARS engine + the evaluation + the written dissertation. **Never cut:** the contribution, the evaluation with metrics, or the writing.

7. Dependencies, assumptions & risks

- **Assumption:** ~35–40 h/week available (full-time dissertation). Scale the plan if less.
- **Dependency:** the pre-build checklist (devices, installs, USB-Ethernet adapters) is done before/at the start of Week 1, or Week 1 slips.

- **Risk — Ryu/Python:** use a Python 3.9/3.10 venv or the os-ken fork if eventlet errors appear.
- **Risk — Factory I/O S7 link:** verify PUT/GET + non-optimized block access on the PLC early (Week 1).
- **Risk — over-building:** the single biggest threat to marks. The gates and ladder exist to counter it; honour the Week-4 writing gate.

Dates are indicative (Wk1 = 6 Jul 2026). Anchor to your actual submission date and confirm the word-count limit in your programme handbook. This plan operationalises the Decision Log; gates map to CC-1/CC-3/CC-4.